

Jean MONLONG  
INSERM IRSD U1220, CHU PURPAN  
31024 Toulouse, FRANCE  
Email: [jean.monlong@inserm.fr](mailto:jean.monlong@inserm.fr)  
Webpage: <https://jmonlong.github.io/>

## Education

---

- 2012 – 2018** [MCGILL UNIVERSITY](#), Montreal, Canada.  
**PhD in the Human Genetics Department** in [Guillaume Bourque's Lab](#).
- 2010 – 2011** [UNIVERSITAT POLITÈCNICA DE CATALUNYA](#), Barcelona, Spain.  
**Master in Statistic and Operations Research** as an exchange student in the Faculty of Mathematics and Statistics.
- 2008 – 2010** [ENSIMAG](#), Grenoble, France.  
**Computer Science and Mathematics** course with specialization in **Bioinformatics**.
- 2006 – 2008** [LYCÉE MICHEL MONTAIGNE](#), Bordeaux, France.  
Preparatory classes for entrance to the Grandes Écoles. Mathematics and Physics.

## Professional Experiences and Projects

---

- 2023 –** [INSTITUT DE RECHERCHE EN SANTÉ DIGESTIVE](#), Toulouse, France.  
Researcher CRCN at [INSERM](#). Genomics research on pangenomes and structural variants.
- 2018 – 2023** [UNIVERSITY OF CALIFORNIA SANTA CRUZ](#), Santa Cruz, USA.  
Postdoctoral researcher with Benedict Paten in the [Computational Genomics Lab](#), analyzing structural variation with pangenomes.
- 2012 – 2018** [BOURQUE LAB](#), [McGill Genome Centre](#), Montreal, Canada.  
PhD project. Implementation of a [CNV detection method](#) for whole-genome sequencing that can deal with repeat-rich regions. Application to large cohorts of normal and disease genomes.
- 2017** [CENTER FOR GENOMIC MEDICINE](#), Kyoto, Japan.  
(3 months) As part of an exchange program, I visited Dr. Matsuda's group and contributed to their ongoing sequencing projects to characterize Japanese genomes.
- 2017** [GENOMICS ENGLAND](#), London, UK.  
(1.5 month) Visiting the bioinformatics team lead by Dr. Rendon, I contributed to the curation of structural variant calls for their genome analysis pipeline.
- 2011 – 2012** [CENTER FOR GENOMIC REGULATION\(CRG\)](#), Barcelona, Spain.  
(1 year) Graduation project in [Roderic Guigó's group](#). Comparison of splicing activity and detection of splicing QTLs from RNA-Seq experiments. Participation in the [Geuvadis](#) and [GTEx](#) projects.
- 2011** [UNIVERSITAT POLITÈCNICA DE CATALUNYA](#), Barcelona, Spain.  
(3 months) Study of the regularization of the generalized canonical correlation analysis.
- 2010** [NEOMADES](#)(Mobile software development), Bidart, France.  
(3 months) Implementation of a Java module of their principal product.
- 2010** [ENSIMAG](#), Grenoble, France.  
(1 month) Breast cancer modelization and Bayesian estimation of the overdiagnosis rate. Collaboration with La Tronche's hospital (Grenoble).

## Expertise

---

R, Java, C, Bash, Python, WDL, Snakemake, AWK, Perl, D3/Javascript/HTML/CSS.  
Git, Linux, High Performance Computing, LaTeX, R/Markdown, Emacs.  
Genomics, Structural Variation, Pangenomics, Transcriptomics, Bioinformatics.

## Funding, Scholarships, and Awards

---

|           |                                                                                                                                              |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------|
| 2023-2027 | <a href="#">Agence National de la Recherche (ANR)</a> Jeunes Chercheurs et Jeunes Chercheuses (JCJC) (335 674 euros).                        |
| 2019-2020 | <a href="#">BioData Catalyst Fellow</a> . National Heart, Lung, and Blood Institute, USA.                                                    |
| 2018      | Department Activity Award. Human Genetics Dept, McGill University.                                                                           |
| 2017      | Student Exchange Support Program. JASSO, Japan.                                                                                              |
| 2017      | Graduate Mobility Award. Graduate and Postdoctoral Studies, McGill University.                                                               |
| 2017      | Oral Presentation, Honorable Mention. Human Genetics Dept's Research Day, McGill University.                                                 |
| 2017      | Excellence Award. Human Genetics Dept, McGill University.                                                                                    |
| 2013-2016 | Graduate Research Enhancement and Travel Award funding attendance to five international conferences. Human Genetics Dept, McGill University. |
| 2015      | Excellence Award. Human Genetics Dept, McGill University.                                                                                    |
| 2014      | Best oral presentation. Human Genetics Dept's Research Day, McGill University.                                                               |

## Publications

---

|      |                                                                                                                                                                                                                                                                                                                                                                                                                   |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2025 | <b>J Monlong</b> , X Chen, H Barseghyan, WJ Rowell, S Negi, N Nokoff, L Mohnach, J Hirsch, C Finlayson, CE Keegan, M Almalvez, SI Berger, I DeDios, B McNulty, A Robertson, K Miga, PW Speiser, B Paten, E Vilain, and EC Délot. Long-read sequencing resolves the clinically relevant CYP21A2 locus, supporting a new clinical test for Congenital Adrenal Hyperplasia. <a href="#">medRxiv</a> , February 2025. |
| 2025 | S Negi, SL Stenton, SI Berger, P Canigiula, B McNulty, I Violich, J Gardner, ..., E Vilain, A O'Donnell-Luria, E Délot, KH Miga, <b>J Monlong</b> , and B Paten. Advancing long-read nanopore genome assembly and accurate variant calling for rare disease detection. <i>The American Journal of Human Genetics</i> , 2025.                                                                                      |
| 2024 | KJ Billingsley, M Meredith, K Daida, P Alvarez Jerez, S Negi, L Malik, ..., <b>J Monlong</b> , ..., M Kolmogorov, M Jain, FJ Sedlazeck, B Paten, and C Blauwendraat. Long-read sequencing of hundreds of diverse brains provides insight into the impact of structural variation on gene expression and DNA methylation. <a href="#">bioRxiv</a> , 2024.                                                          |
| 2024 | AM Novak, D Chung, G Hickey, S Djebali, TT Yokoyama, E Garrison, G Narzisi, B Paten, and <b>J Monlong</b> . Efficient indexing and querying of annotations in a pangenome graph. <a href="#">bioRxiv</a> , 2024.                                                                                                                                                                                                  |
| 2024 | J Sirén, P Eskandar, MT Ungaro, G Hickey, JM Eizenga, AM Novak, X Chang, PC Chang, M Kolmogorov, A Carroll, <b>J Monlong</b> , and B Paten. Personalized pangenome references. <i>Nature Methods</i> , 2024.                                                                                                                                                                                                      |
| 2024 | J Yan, <b>J Monlong</b> , C Cougoule, S Lacroix-Lamandé, and A Wiedemann. Mapping the scientific output of organoids for animal and human modeling infectious diseases: a bibliometric assessment. <i>Veterinary Research</i> , 55, 2024.                                                                                                                                                                         |
| 2024 | DJ Taylor, JM Eizenga, Q Li, A Das, KM Jenike, EE Kenny, KH Miga, <b>J Monlong</b> , RC McCoy, B Paten, and MC Schatz. Beyond the Human Genome Project: The Age of Complete Human Genome Sequences and Pangenome References. <i>Annual Review of Genomics and Human Genetics</i> , 2024.                                                                                                                          |

- 2024 R Lorig-Roach, M Meredith, **J Monlong**, ..., P Carnevali, K Miga, and B Paten. Phased nanopore assembly with Shasta and modular graph phasing with GFase. *Genome Research*, 2024.
- 2024 E Kuzmin, TM Baker, T Lesluyes, **J Monlong**, KT Abe, PP Coelho, M Schwartz, ..., J Ragoussis, G Bourque, P Van Loo, and M Park. Evolution of chromosome-arm aberrations in breast cancer through genetic network rewiring. *Cell Reports*, 2024.
- 2023 M Kolmogorov, KJ Billingsley, M Mastoras, M Meredith, **J Monlong**, ..., C Blauwendraat, and B Paten. Scalable Nanopore sequencing of human genomes provides a comprehensive view of haplotype-resolved variation and methylation. *Nature Methods*, 2023.
- 2023 G Hickey\*, **J Monlong\***, J Ebler, AM Novak, JM Eizenga, Y Gao, Human Pangenome Reference Consortium, H Li, and B Paten. Pangenome graph construction from genome alignments with Minigraph-Cactus. *Nature Biotechnology*, 2023.
- 2023 W Liao, M Asri, J Ebler, D Doerr, M Haukness, G Hickey, S Lu, JK Lucas, **J Monlong**, ..., G Bourque, MJP Chaisson, P Flicek, AM Phillippy, JM Zook, EE Eichler, D Haussler, T Wang, ED Jarvis, KH Miga, E Garrison, T Marschall, IM Hall, H Li, and B Paten. A draft human pangenome reference. *Nature*, 2023.
- 2022 CP Couturier, J Nadaf, Z Li, S Baig, G Riva, P Le, DJ Kloosterman, **J Monlong**, A Nkili Meyong, R Allache, T Degenhard, M Al-Rashid, M Guiot, G Bourque, J Ragoussis, L Akkari, FJ Quintana, and K Petrecca. Glioblastoma scRNAseq Shows Treatment-induced, Immune-dependent Rise In Mesenchymal Cancer Cells, and Structural Variants in Distal Neural Stem Cells. *Neuro-Oncology*, 2022.
- 2022 SD Goenka, JE Gorzynski, K Shafin, DG Fisk, T Pesout, TD Jensen, **J Monlong**, P Chang, G Baid, ..., B Paten, and EA Ashley. Accelerated identification of disease-causing variants with ultra-rapid nanopore genome sequencing. *Nature Biotechnology*, 2022.
- 2022 JE Gorzynski, SD Goenka, K Shafin, TD Jensen, DG Fisk, ME Grove, E Spiteri, T Pesout, **J Monlong**, JA Bernstein, ..., B Paten, and EA Ashley. Ultra-Rapid Nanopore Whole Genome Genetic Diagnosis of Dilated Cardiomyopathy in an Adolescent With Cardiogenic Shock. *Circulation: Genomic and Precision Medicine*, 2022.
- 2022 JE Gorzynski, SD Goenka, K Shafin, TD Jensen, DG Fisk, ME Grove, E Spiteri, T Pesout, **J Monlong**, ..., B Paten, and EA Ashley. Ultrarapid Nanopore Genome Sequencing in a Critical Care Setting. *New England Journal of Medicine*, 2022.
- 2021 J Sirén\*, **J Monlong\***, X Chang\*, AM Novak\*, JM Eizenga\*, C Markello, JA Sibbesen, G Hickey, P Chang, A Carroll, N Gupta, S Gabriel, TW Blackwell, A Ratan, KD Taylor, SS Rich, JI Rotter, D Haussler, E Garrison, and B Paten. Pangenomics enables genotyping of known structural variants in 5202 diverse genomes. *Science*, 2021.
- 2021 E Kuzmin, **J Monlong**, C Martinez, H Kuasne, CL Kleinman, J Ragoussis, G Bourque, and M Park. Inferring Copy Number from Triple-Negative Breast Cancer Patient Derived Xenograft scRNAseq Data Using scCNA. In *Mapping Genetic Interactions*, volume 2381, pages 285–303. Springer US, 2021.

- 2021** M Dankner, M Caron, T Al-Saadi, W Yu, V Ouellet, R Ezzeddine, SM Maritan, MG Annis, PU Le, J Nadaf, NS Neubarth, P Savage, D Zuo, CP Couturier, **J Monlong**, H Djambazian, H Altoukhi, G Bourque, J Ragoussis, RJ Diaz, M Park, M Guiot, S Lam, K Petrecca, and PM Siegel. Invasive growth associated with cold-inducible RNA-binding protein expression drives recurrence of surgically resected brain metastases. *Neuro-Oncology*, 2021.
- 2020** Y Wang, JYP Park, A Pacis, RE Denroche, GH Jang, A Zhang, A Cuggia, C Domecq, **J Monlong**, ..., and G Zogopoulos. A Preclinical Trial and Molecularly Annotated Patient Cohort Identify Predictive Biomarkers in Homologous Recombination-deficient Pancreatic Cancer. *Clinical Cancer Research*, 2020.
- 2020** CP Couturier, S Ayyadhury, PU Le, J Nadaf, **J Monlong**, ..., J Antel, G Bourque, J Ragoussis, and K Petrecca. Single-cell RNA-seq reveals that glioblastoma recapitulates a normal neurodevelopmental hierarchy. *Nature Communications*, 2020.
- 2020** K Shafin, T Pesout, R Lorig-Roach, M Haukness, HE Olsen, ..., **J Monlong**, ..., KH Miga, P Carnevali, M Jain, and B Paten. Nanopore sequencing and the Shasta toolkit enable efficient de novo assembly of eleven human genomes. *Nature Biotechnology*, 2020.
- 2020** G Hickey\*, D Heller\*, **J Monlong\***, JA Sibbesen, J Siren, J Eizenga, ET Dawson, E Garrison, AM Novak, and B Paten. Genotyping structural variants in pangenome graphs using the vg toolkit. *Genome Biology*, 2020.
- 2020** B Rodriguez-Martin, EG Alvarez, A Baez-Ortega, ..., **J Monlong**, ..., HH Kazazian, KH Burns, PJ Campbell, and JMC Tubio. Pan-cancer analysis of whole genomes identifies driver rearrangements promoted by LINE-1 retrotransposition. *Nature Genetics*, 2020.
- 2019** S Jessa, A Blanchet-Cohen, B Krug, M Vladoiu, M Coutelier, D Faury, B Poreau, N De Jay, S Hébert, **J Monlong**, ..., G Bourque, J Ragoussis, L Garzia, MD Taylor, N Jabado, and CL Kleinman. Stalled developmental programs at the root of pediatric brain tumors. *Nature Genetics*, 2019.
- 2019** J Grajcarek, **J Monlong**, Y Nishinaka-Arai, M Nakamura, M Nagai, S Matsuo, D Loughheed, H Sakurai, MK Saito, G Bourque, and K Woltjen. Genome-wide microhomologies enable precise template-free editing of biologically relevant deletion mutations. *Nature Communications*, 2019.
- 2018** **J Monlong**, P Cossette, C Meloche, G Rouleau, SL Girard, and G Bourque. Human copy number variants are enriched in regions of low mappability. *Nucleic Acids Research*, 2018.
- 2018** **J Monlong\***, SL Girard\*, ..., JL Michaud, G Rouleau, BA Minassian, G Bourque, and P Cossette. Global characterization of copy number variants in epilepsy patients from whole genome sequencing. *PLOS Genetics*, 2018.
- 2017** M Arseneault\*, **J Monlong\***, ..., M Lathrop, G Bourque, and Y Riazalhosseini. Loss of chromosome Y leads to down regulation of KDM5D and KDM6C epigenetic modifiers in clear cell renal cell carcinoma. *Scientific Reports*, 2017.

- 2017 F Hamdan, CT Myers, P Cossette, P Lemay, D Spiegelman, A Dionne-Laporte, C Nassif, O Diallo, **J Monlong**, ..., BA Minassian, and JL Michaud. High Rate of Recurrent De Novo Mutations in Developmental and Epileptic Encephalopathies. *American journal of human genetics*, 2017.
- 2017 **The GTEx Consortium**. Genetic effects on gene expression across human tissues. *Nature*, 2017.
- 2017 A Saha, et al., and **The GTEx Consortium**. Co-expression networks reveal the tissue-specific regulation of transcription and splicing. *Genome research*, 2017.
- 2017 F Yang, et al., and **The GTEx Consortium**. Identifying cis-mediators for trans-eQTLs across many human tissues using genomic mediation analysis. *Genome research*, 2017.
- 2017 X Li, et al., and **The GTEx Consortium**. The impact of rare variation on gene expression across tissues. *Nature*, 2017.
- 2017 MH Tan, et al., and **The GTEx Consortium**. Dynamic landscape and regulation of RNA editing in mammals. *Nature*, 2017.
- 2017 T Tukiainen, et al., and **The GTEx Consortium**. Landscape of X chromosome inactivation across human tissues. *Nature*, 2017.
- 2015 M Melé, PG Ferreira, F Reverter, DS DeLuca, **J Monlong**, M Sammeth, ..., and The GTEx Consortium. The human transcriptome across tissues and individuals. *Science*, 2015.
- 2015 **The GTEx Consortium**. The Genotype-Tissue Expression (GTEx) pilot analysis: Multitissue gene regulation in humans. *Science*, 2015.
- 2015 D Pervouchine, S Djebali, A Breschi, CA Davis, PP Barja, A Dobin, A Tanzer, J Lagarde, C Zaleski, L See, M Fastuca, J Drenkow, H Wang, G Bussotti, B Pei, S Balasubramanian, **J Monlong**, ..., C Notredame, R Guigo, and TR Gingeras. Enhanced transcriptome maps from multiple mouse tissues reveal evolutionary constraint in gene expression. *Nature Communications*, 2015.
- 2014 **J Monlong**, M Calvo, PG Ferreira, and R Guigó. Identification of genetic variants associated with alternative splicing using sQTLseeker. *Nature Communications*, 2014.
- 2014 PG Ferreira, P Jares, D Rico, G Gomez-Lopez, A Martinez-Trillos, N Villamor, S Ecker, A Gonzalez-Perez, DG Knowles, **J Monlong**, ..., E Campo, and R Guigo. Transcriptome characterization by RNA sequencing identifies a major molecular and clinical subdivision in chronic lymphocytic leukemia. *Genome research*, 2014.
- 2014 L Greger, J Su, J Rung, PG Ferreira, **The Geuvadis Consortium**, T Lappalainen, ET Dermitzakis, and A Brazma. Tandem RNA chimeras contribute to transcriptome diversity in human population and are associated with intronic genetic variants. *PloS one*, 2014.
- 2013 T Lappalainen, M Sammeth\*, MR Friedlander\*, PA t Hoen\*, **J Monlong\***, MA Rivas\*, and et al. Transcriptome and genome sequencing uncovers functional variation in humans. *Nature*, 2013.

**2013**

PC t Hoen, et al., and **The Geuvadis Consortium**. Reproducibility of high-throughput mRNA and small RNA sequencing across laboratories. *Nature biotechnology*, 2013.

**2013**

**The GTEx Consortium**. The Genotype-Tissue Expression (GTEx) project. *Nat Genet*, 2013.

**2011**

A Seigneurin, O Francois, J Labarere, P Oudeville, **J Monlong**, and M Colonna. Overdiagnosis from non-progressive cancer detected by screening mammography: stochastic simulation study with calibration to population based registry data. *BMJ*, 2011.

## Selected Scientific Presentations

---

- 2024** SWITZERLAND, Ascona.  
**Oral** presentation, *International Genome Graph Symposium*.
- 2024** FRANCE, Toulouse.  
**Oral** presentation, *Journées Ouvertes en Biologie, Informatique, et Mathématiques*.
- 2020** NY, USA, New York City.  
**Oral** presentation, *GSP-TOPMed Analysis Workshop*.
- 2019** NY, USA, Cold Spring Harbor Laboratory.  
**Oral** presentation, *Genome Informatics* conference.
- 2017** QUEBEC, CANADA, Quebec.  
**Oral** presentation, *6th Annual Canadian Human and Statistical Genetics Meeting*.
- 2016** JAPAN, Kyoto.  
**Oral** presentation, *13th International Congress of Human Genetics*.
- 2015** BC, CANADA, Vancouver.  
**Oral** presentation, *4rd Annual Canadian Human and Statistical Genetics Meeting*.
- 2014** UK, Cambridge.  
**Oral** presentation, *Genome Informatics* conference.

## Other Scientific Presentations

---

- 2020** NY, USA, New York City.  
**Oral** presentation, *GSP-TOPMed Analysis Workshop*.
- 2019** TX, USA, Houston.  
**Oral** presentation, *GRC-GIAB Workshop* at ASHG.
- 2019** CA, USA, UC Davis.  
**Oral** presentation, *Northern California Computational Biology Symposium*.
- 2017** QUEBEC, CANADA, Montreal.  
**Oral** presentation, Human Genetics Dept. Research Day.
- 2016** SPAIN, Barcelona.  
**Poster** presentation, *European Society of Human Genetics* conference. Best Poster Award candidate.
- 2015** SCOTLAND, Glasgow.  
**Poster** presentation, *European Society of Human Genetics* conference.
- 2014** QUEBEC, CANADA, Montreal.  
**Oral** presentation, Human Genetics Dept. Research Day.
- 2014** BRITISH COLUMBIA, CANADA, Victoria.  
**Poster** presentation at the *3rd Annual Canadian Human and Statistical Genetics Meeting*.
- 2014** NEW MEXICO, USA, Santa Fe.  
**Poster** presentation at the *Mobile Genetic Elements and Genome Evolution* Keystone conference.
- 2013** QUEBEC, CANADA, Esterel.  
**Poster** presentation at the *2nd Annual Canadian Human and Statistical Genetics Meeting*.
- 2013** MONTANA, USA, Big Sky.  
**Poster** presentation at the *Mobile DNA in Mammalian Genomes* FASEB conference.
- 2011** SPAIN, Barcelona.  
**Poster** presentation at the *XIIIth Spanish Biometry Conference and 3rd Ibero-American Biometry Meeting*.

## Other Activities

---

|           |                                                                                                                                                                 |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2024      | ROME, ITALY.<br>Instructor at the <i>HPRC workshop</i> during the Human Genome Meeting (HUGO).                                                                  |
| 2023      | SALT LAKE CITY, USA.<br>Instructor at the <i>HPRC workshop</i> during the ACMG Annual Clinical Genetics Meeting.                                                |
| 2019      | HOUSTON, USA.<br>Instructor at the <i>GRC-GIAB workshop</i> during the American Society of Human Genetics (ASHG) Annual Meeting.                                |
| 2019-2023 | UNIVERSITY OF CALIFORNIA, SANTA CRUZ.<br>Representative for the UCSC STEM Postdoc Association. Co-chair of the 2020 USPA Symposium.                             |
| 2018      | IRCM, MONTREAL.<br>Gave a <a href="#">short seminar about R Markdown</a> at the Montreal Bioinformatics User Group.                                             |
| 2013-2018 | MCGILL UNIVERSITY.<br>Organizer and instructor of eight full-day <i>Introduction to R and Bioinformatics</i> workshops for Human Genetics graduate students.    |
| 2014-2017 | MCGILL UNIVERSITY.<br>VP Communications at the <a href="#">Human Genetics Student Society</a> .                                                                 |
| 2017      | UNIVERSITÉ DE MONTREAL.<br>Participated to a Bio-Hackaton. Our 24h project was to detect extending filaments from 3D cell imaging.                              |
| 2017      | IRCM, MONTREAL.<br>Gave a <a href="#">short seminar about the Gviz package</a> at the Montreal Bioinformatics User Group.                                       |
| 2016      | KYOTO, JAPAN.<br>Instructor at the <i>4th Kyoto Course and Symposium on Bioinformatics for Next Generation Sequencing with Applications in Human Genetics</i> . |
| 2015      | UNIVERSITÉ DE MONTREAL.<br>Participated to a Bio-Hackaton. Our 24h project was to develop interactive visualization of gene expression in cancer samples.       |
| 2014      | MCGILL UNIVERSITY.<br>Teaching Assistant for the <i>Advanced Statistical Concepts in Genetic and Genomic Analysis</i> course.                                   |
| 2014      | MCGILL UNIVERSITY.<br>Organizer of a four-days <i>R and Bioinformatics</i> workshop for System Biology graduate students.                                       |